Sagar Kriplani

Mechanical Design & Development Engineer - M.Sc. Scientific Instrumentation

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Work Authorization: EU Blue Card (Unrestricted, Available Immediately)

PROFESSIONAL SUMMARY

Results-driven Mechanical Design & Development Engineer (M.Sc., Grade 1.8) with over 3 years of hands-on experience in mechanical design, mechatronic system integration, and thermomechanical FEA simulation within semiconductor equipment (vacuum/thermal subsystems, FOUP/EUV Pod cleaners) and custom machinery (Sondermaschinenbau). Domain expertise in semiconductor equipment performance (wafer thermal chucking, high-vacuum beamline mechanics, equipment reliability). Proven concept-to-release responsibility utilizing Autodesk Inventor 2024, PTC Creo, and Siemens NX, DFM/DFA, GD&T tolerance stacks, and quantifiable system optimization (up to 70% component cost reduction, 50% energy savings). Actively advancing design capabilities in SolidWorks AI workflow automation and generative CAD integration. Business fluent in English (C1) and professional German (B2).

WORK EXPERIENCE

Mechanical Design & Development Engineer - NeSt GmbH Sondermaschinenbau & Anlagenbau, Herbrechtingen 09/2025 - 09/2026

- **Platform Engineering & Production Drawings:** Managed full concept-to-release design cycles for over 8 custom machine platforms, generating approx. 200 production drawings per month compliant with DIN/ISO standards.
- **DFM/DFA & Sheet Metal Assemblies:** Engineered complex machine enclosures, sheet metal enclosures (Trumpf unfolding), and welded assemblies strictly applying DFM/DFA principles.
- **Cost & Weight Optimization:** Achieved significant cost reductions (up to -70%) and weight savings (-60%) through component standardization, material substitution, and FMEA-guided design adjustments.
- **FEA Simulation & Tolerance Analysis:** Performed structural and thermal FEA simulations via ANSYS alongside GD&T tolerance stack calculations to validate mechanical integrity.
- **PDM & ERP Integration:** Maintained revision-controlled CAD assemblies and Bills of Materials (BOM) in PDM and SAGE ERP; coordinated technical interfaces across R&D, assembly, and external suppliers.

Working Student (Mechanical Design - Specialized Equipment) - Brooks Automation (Deutschland) GmbH, Steißlingen 02/2024 - 06/2024

- **Semiconductor Vacuum Equipment:** Designed high-precision vacuum chamber modules and mechatronic assemblies for FOUP cleaning systems used in cleanroom class 1 µm semiconductor manufacturing according to SEMI standards.
- **Optothermal Re-Engineering:** Re-engineered IR quartz heater mounting brackets, reducing thermal shadowing by 80% while achieving 3x weight reduction; validated via FLIR thermography and optical metrology.
- **PLM & CAD Documentation:** Maintained BOM structures, handled engineering change notices (ECN), and generated technical documentation in Agile PLM and Autodesk Inventor.

Mechanical Design & System Integration Intern - Brooks Automation (Deutschland) GmbH, Steißlingen 01/2023 - 06/2023

- **Process Air Unit (PAU) Redesign:** Led the redesign and thermodynamic layout of the Process Air Unit (PAU) for EUV Pod Cleaners from concept through physical prototype validation.
- **Energy & Footprint Reduction:** Reduced electrical heating power from 48 kW to 18 kW (~50% energy reduction), cut manufacturing costs by 21%, decreased installation footprint by 34%, and lowered pressure drop by 90% (500 to 50 mbar).
- **Fluid Control & Standards:** Selected stainless steel grades (304/316Ti) and integrated fluid control components adhering to ASME B31.3 and SEMI S8 standards.
- **Cleanroom Testing:** Conducted cleanroom laboratory testing; performance formally evaluated as "consistently to our complete satisfaction" (stets zur vollsten Zufriedenheit).

Design & Development Engineer - Ravi Brass Industries, Jamnagar, India 08/2020 - 08/2021

- **Precision Electrical Connectors:** Engineered precision electrical connectors and micro-components using Creo Parametric per DIN/IEC standards.
- **Tooling & Mold Design:** Designed 4 custom forming and injection molding tools for high-volume series production exceeding 100,000 components/month; reduced scrap rates by 20% through tolerance optimization.

KEY ENGINEERING PROJECTS

Radiant Heater & LED for Vacuum Polymer Heating Master's Thesis (Grade: 1.3) — Brooks Automation / EAH Jena

- Executed thermal system design, coupled ANSYS FEA simulation, and experimental validation in high vacuum for FOUP cleaner vacuum chambers, achieving a temperature uniformity of ±2 °C.

Automated Pick-and-Place Robotic Arm Kinematics Advanced 3D CAD (Grade: 1.3) — EAH Jena

- Engineered a high-precision robotic arm mechanism (<50 µm positioning accuracy) in Autodesk Inventor, including static structural verification and stress analysis.

Numerical Modeling of Heat & Mass Transfer Bachelor's Thesis (Grade: 1.0) — GTU India

- Performed numerical modeling and CFD simulation of heat and mass transfer during frost growth processes using MATLAB for cryogenic and HVAC applications.

EDUCATION

Master of Science (M.Sc.) - Scientific Instrumentation Final Grade: 1.8 (ECTS A) — 10/2021 - 03/2025

Ernst-Abbe-Hochschule Jena, Germany

Key Modules: Advanced 3D CAD Design (Grade 1.3), FEM & Simulation (Grade 2.3), Mechatronics, Precision Engineering, Materials Diagnostics.

Bachelor of Engineering (B.E.) - Mechanical Engineering Final Grade: 1.9 (CGPA: 8.14 / 10.00) — 08/2017 - 08/2020

Gujarat Technological University (A. D. Patel Institute of Technology), India

Diploma in Mechanical Engineering Final Grade: 1.2 (CGPA: 9.50 / 10.00) — 08/2014 - 06/2017

Gujarat Technological University (Government Polytechnic Jamnagar), India

CERTIFICATIONS & CONTINUOUS DEVELOPMENT

Siemens NX: Basic to NC Programming	Siemens Centre of Excellence
ELEATION: ANSYS Basic to Professional Training (FEA Structural & Thermal, Score: 325/325)	ELEATION
Semiconductor Innovation: Advanced Technology & Geopolitics	Tech Diplomacy Academy
SolidWorks AI Workflow Automation: Generative CAD & Parametric Automation	10/2026 - Present

TECHNICAL SKILLS & COMPETENCIES

CAD Systems:	SolidWorks (AI workflows & parametric), Autodesk Inventor 2024 (Expert), PTC Creo, Siemens NX, AutoCAD
FEA & Simulation:	ANSYS Workbench (Structural & Thermal), CFD fundamentals, MATLAB & Simulink, GD&T (DIN/ISO 1101)
Semiconductor & Vacuum:	High-Vacuum & Differential Pumping, Wafer Thermal Management, Cleanroom Class 1 µm, SEMI S2/S8, ASME B31.3
Methodology & Manufacturing:	Custom Machinery (Sondermaschinenbau), Sheet Metal (Trumpf), Welded Assemblies, DFM/DFA, FMEA, ECN
PLM, ERP & Software:	SAGE ERP, Agile PLM, Autodesk Vault / PDM, Python (Basics), MS Office 365
Languages:	**English:** C1 (Full Professional) | **German:** B2 (Professional Working) | **Hindi, Gujarati, Sindhi:** Native